

SIT720: Machine Learning

Distinction Task 8.1D Machine Learning Mini Project

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Introduction

The objective of this project is to develop a machine learning-based property sale price prediction system using real estate data. The project focuses on applying regression techniques to estimate property prices based on important property characteristics, including location, property type, number of bedrooms, bathrooms, car spaces, building size, and other relevant features.

A complete machine learning workflow was implemented, including dataset preparation, data preprocessing, feature engineering, model development, performance evaluation, and deployment. Multiple regression models were developed and compared to identify the best-performing approach for property price prediction.

In addition to automated machine learning prediction, this project also compares predictions generated by a large language model (LLM) and human judgement using selected properties from a held-out test dataset. This comparison provides insight into the strengths and limitations of different approaches for property valuation.

Finally, the selected machine learning model was deployed through a Streamlit-based web application. The application allows users to enter property information and receive an estimated sale price generated by the trained model. The project concludes with a reflection on the challenges, ethical considerations, and possible improvements of automated property valuation systems.

Project Repository:

All project resources, including the collected Sydney housing dataset, data preprocessing and machine learning notebook, deployed Streamlit application source code, trained model, and supporting files are available through the GitHub repository:

<https://github.com/nafishasnat2081/SIT720-Property-Price-Prediction>

The repository has been organised to support reproducibility and provides access to the dataset, implementation details, and deployment resources required to reproduce the results presented in this report.

1. Part 1: Problem Definition and Data Collection

1.1 Problem Definition

Housing price prediction is an important real-world machine learning problem that aims to estimate the sale price of a property based on its characteristics and surrounding market conditions. Accurate property valuation can support buyers, sellers, real estate agencies, and financial institutions by providing data-driven estimates of property values.

The problem is formulated as a supervised regression task where historical property characteristics are used to learn relationships between input features and continuous sale price values.

In this project, the objective is to develop a machine learning model that predicts the sale price of residential properties in selected Sydney suburbs. The prediction target is the property sale price in Australian Dollars (AUD). The model uses different property characteristics, including location, property type, number of bedrooms, number of bathrooms, car spaces, building size, and market-related features to learn patterns associated with housing values.

1.2 Suburb Selection

Three Sydney housing markets were selected for this project:

- **Surry Hills**
- **Greater Parramatta**
- **Campbelltown**

These suburbs were selected because they represent substantially different housing markets. Surry Hills represents an inner-city suburb with higher property values, strong accessibility, and a greater proportion of apartment-based housing. Greater Parramatta represents a rapidly developing metropolitan centre with diverse residential properties. Campbelltown represents a more suburban housing market with comparatively different property characteristics and price ranges.

The differences in location, accessibility, property type distribution, and market conditions provide a suitable basis for developing and evaluating a housing price prediction model.

1.3 Data Collection

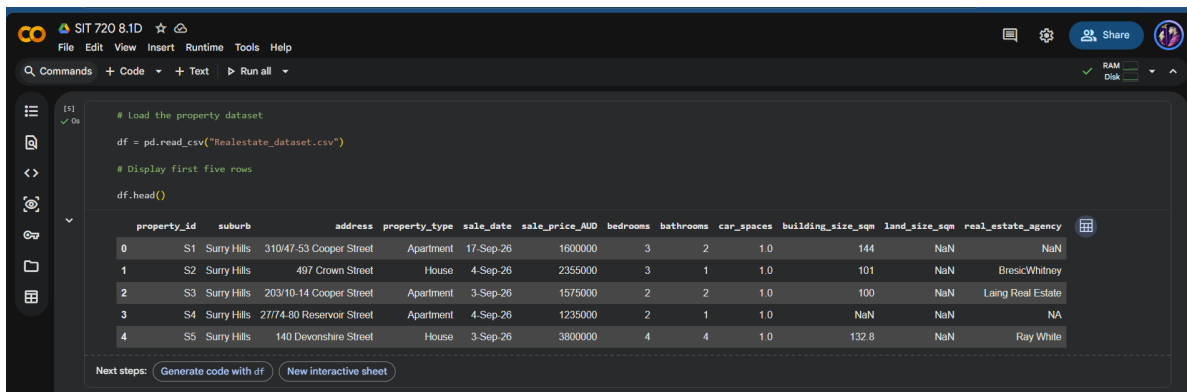
A dataset was manually constructed using publicly available property information from real estate listing sources. A total of 139 sold property records were collected across the three selected Sydney suburbs. The dataset included 44 properties from Surry Hills, 49 properties from Greater Parramatta, and 46 properties from Campbelltown, ensuring that each selected suburb exceeded the minimum requirement of 30 properties.

For each property, the collected information included the sale price (prediction target) and a range of property features that may influence housing value. The collected attributes include:

- Property ID
- Suburb
- Address

- Property type
- Sale date
- Sale price (AUD)
- Number of bedrooms
- Number of bathrooms
- Number of car spaces
- Building size (sqm)
- Real estate agency
- Market region

The initial dataset inspection was performed to understand the available variables, number of records, and overall dataset structure.

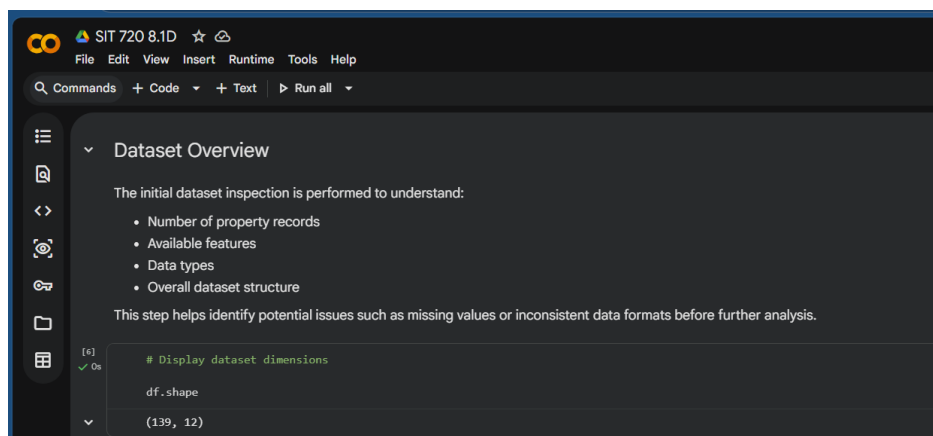


	property_id	suburb	address	property_type	sale_date	sale_price_AUD	bedrooms	bathrooms	car_spaces	building_size_sqm	land_size_sqm	real_estate_agency
0	S1	Surry Hills	310/47-53 Cooper Street	Apartment	17-Sep-26	1600000	3	2	1.0	144	NaN	NaN
1	S2	Surry Hills	497 Crown Street	House	4-Sep-26	2355000	3	1	1.0	101	NaN	BresicWhitney
2	S3	Surry Hills	203/10-14 Cooper Street	Apartment	3-Sep-26	1575000	2	2	1.0	100	NaN	Laing Real Estate
3	S4	Surry Hills	27/74-80 Reservoir Street	Apartment	4-Sep-26	1235000	2	1	1.0	NaN	NaN	NA
4	S5	Surry Hills	140 Devonshire Street	House	3-Sep-26	3800000	4	4	1.0	132.8	NaN	Ray White

Figure 1: Initial view of the collected property dataset.

1.4 Dataset Overview

The dataset dimensions were examined before further analysis. The collected dataset contained 139 property records with 12 variables.



Dataset Overview

The initial dataset inspection is performed to understand:

- Number of property records
- Available features
- Data types
- Overall dataset structure

This step helps identify potential issues such as missing values or inconsistent data formats before further analysis.

```
# Display dataset dimensions
df.shape
```

(139, 12)

Figure 2: Dataset dimensions showing the number of collected property records and features.

The distribution of collected properties across the three selected markets was also examined. The dataset contained properties from all three suburbs, ensuring representation of different housing markets.

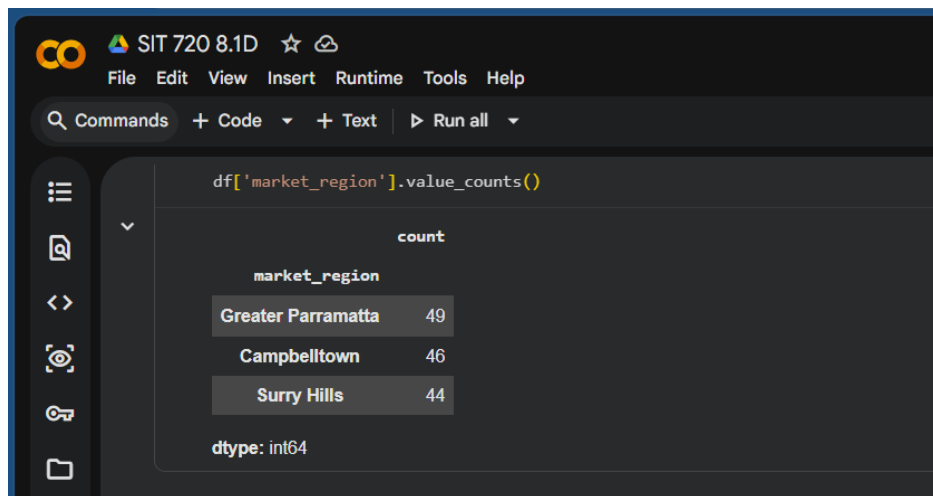


Figure 3: Distribution of collected properties across selected Sydney markets.

1.5 Data Quality Assessment

Before model development, the dataset was analysed for missing information and inconsistent data formats. Missing value analysis identified incomplete attributes such as building size, land size, and real estate agency information.

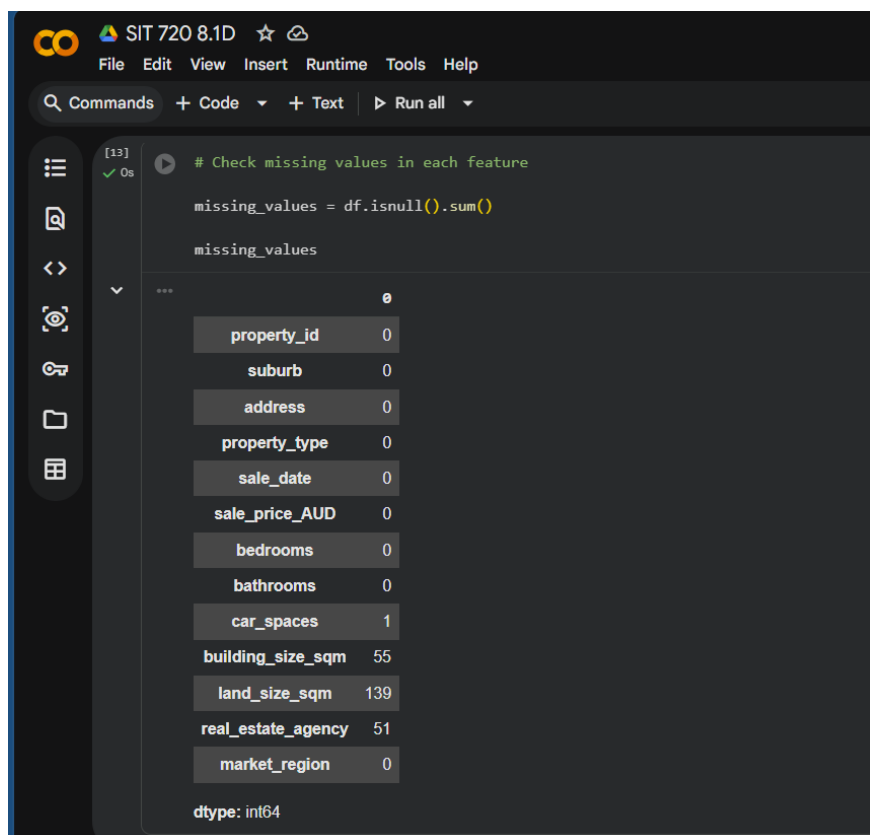
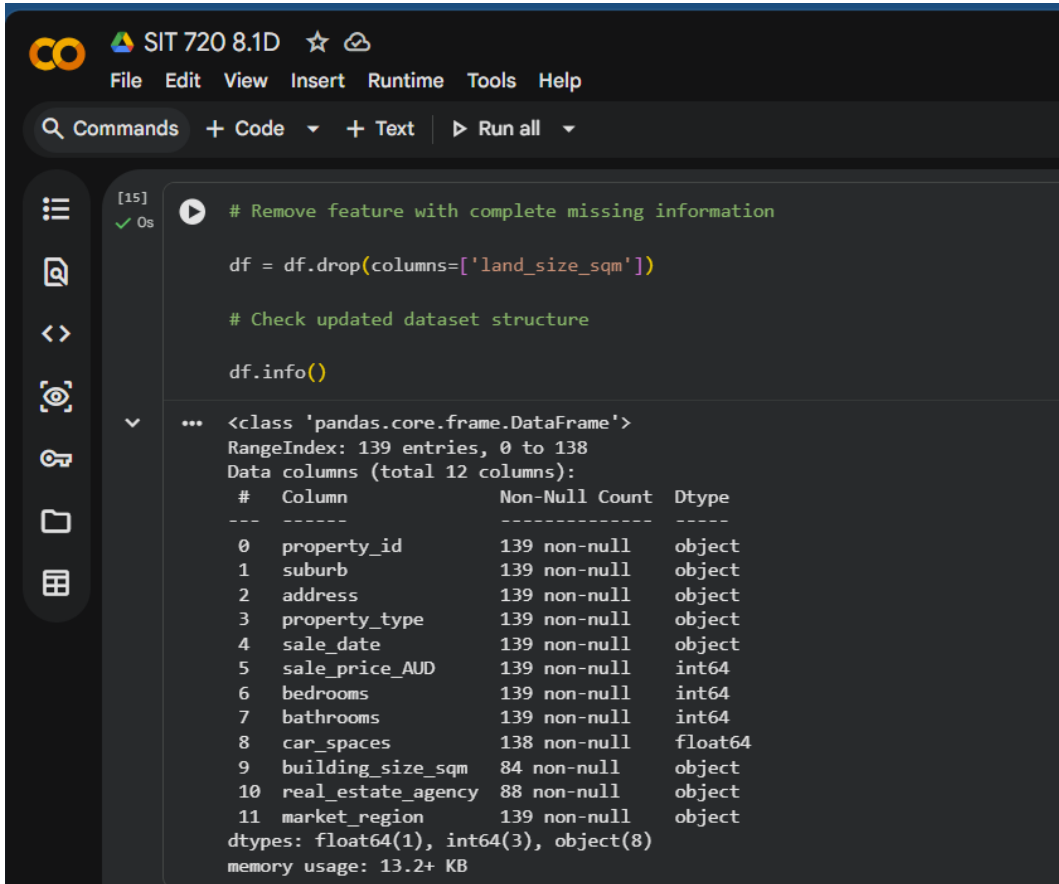


Figure 4: Missing value analysis for collected property features.

Missing value analysis showed that land size information was unavailable for all collected properties, while

building size and real estate agency information contained 55 and 51 missing records respectively.

The dataset structure and variable types were then examined to determine appropriate preprocessing requirements. Numerical variables and categorical variables were identified for later transformation and machine learning model development.



```
[15] ✓ Os # Remove feature with complete missing information

df = df.drop(columns=['land_size_sqm'])

# Check updated dataset structure

df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 139 entries, 0 to 138
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   property_id           139 non-null    object
1   suburb                139 non-null    object
2   address               139 non-null    object
3   property_type         139 non-null    object
4   sale_date             139 non-null    object
5   sale_price_AUD        139 non-null    int64
6   bedrooms              139 non-null    int64
7   bathrooms             139 non-null    int64
8   car_spaces            138 non-null    float64
9   building_size_sqm     84 non-null     object
10  real_estate_agency     88 non-null     object
11  market_region         139 non-null    object
dtypes: float64(1), int64(3), object(8)
memory usage: 13.2+ KB
```

Figure 5: Dataset structure showing feature names, non-null counts, and data types after initial preprocessing.

1.6 Challenges and Limitations

During data collection, several challenges were identified. Since the dataset was manually collected, the number of available properties was limited compared with large commercial property databases. Some important property attributes were unavailable for all records, resulting in missing values.

Additionally, housing prices are influenced by many external factors that were not included in the dataset, such as property condition, renovation history, proximity to amenities, transport accessibility, local economic conditions, and market trends. Therefore, predictions generated from this dataset may contain uncertainty.

Potential sources of bias may also exist because the collected properties represent selected suburbs and available online listings rather than the complete Sydney housing market. Consequently, the developed model should be interpreted as a predictive tool for the selected dataset rather than a universal property valuation system.

2. Part 2: Data Understanding and Feature Engineering

Before developing predictive models, exploratory data analysis (EDA) was performed to understand the characteristics of the collected housing dataset. The analysis focused on the distribution of housing prices, differences between selected Sydney housing markets, temporal variations, and identification of unusual observations. Visualisations were used to support the interpretation of important patterns within the dataset.

2.1 Exploratory Data Analysis of Housing Prices

The distribution of property sale prices was initially examined using descriptive statistics. The dataset contains 139 property records collected from three Sydney housing markets: Surry Hills, Campbelltown, and Greater Parramatta. The analysis was performed before model development to understand underlying patterns and potential data issues. The sale prices range from AUD 360,000 to AUD 3,800,000, with an average value of approximately AUD 1.05 million.

The descriptive statistics indicate substantial variation in housing prices. The difference between the median and mean values suggests the presence of several high-value properties that increase the average sale price. These observations are important for predictive modelling because extreme property values may influence model performance.

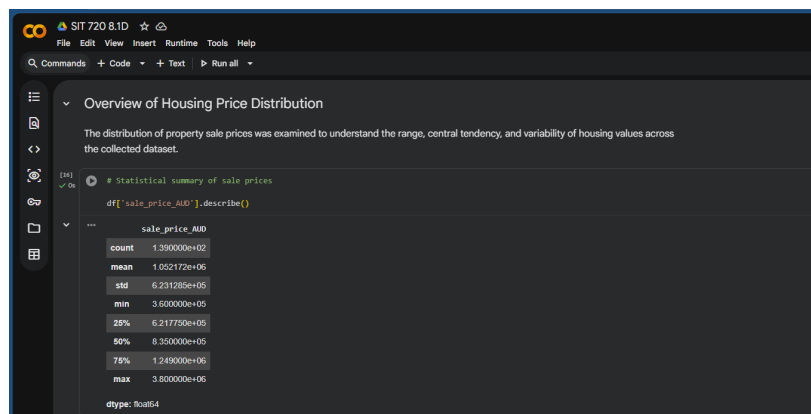


Figure 6: Statistical summary of property sale prices

2.2 Distribution of Property Sale Prices

A histogram was generated to examine the overall distribution of housing prices. The results show a positively skewed distribution, where most properties are concentrated within the lower to middle price ranges while fewer high-value properties extend towards the upper end.

The presence of expensive properties above AUD 2 million reflects realistic differences in location, property type, and property characteristics. These observations were retained rather than removed because they may represent genuine high-end properties.



Figure 7: Distribution of property sale prices

2.3 Outlier Analysis

A boxplot was used to identify potential outliers in the sale price variable. Several observations were located above the upper whisker, indicating properties with substantially higher sale prices compared with the majority of the dataset.

These values were not removed because they may represent genuine luxury properties rather than incorrect records. In real estate markets, extreme prices can occur due to factors such as location, property size, and unique property characteristics.



Figure 8: Boxplot of property sale prices

2.4 Comparison of Different Housing Markets

The collected dataset represents three different Sydney housing markets. Average and median sale prices were compared to identify differences between these markets.

The comparison shows that Surry Hills has the highest average and median property prices, while Campbelltown contains comparatively lower-priced properties. Greater Parramatta represents an intermediate market. These differences indicate that location plays an important role in determining housing value.

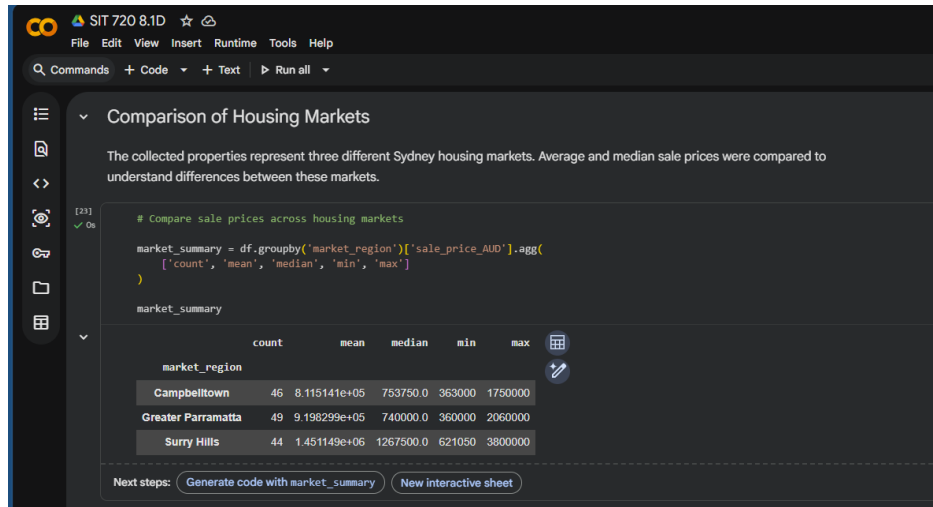


Figure 9: Comparison of housing markets based on sale prices

2.5 Sale Price Trends Over Time

The sale date information was analysed to identify possible temporal variations in property prices. The average sale price was calculated for each sale date.

Although the dataset covers a relatively short collection period, fluctuations in average prices were observed. These variations may occur due to differences in property characteristics and the timing of individual property transactions. However, because the dataset covers a relatively short period, long-term market trends cannot be inferred.

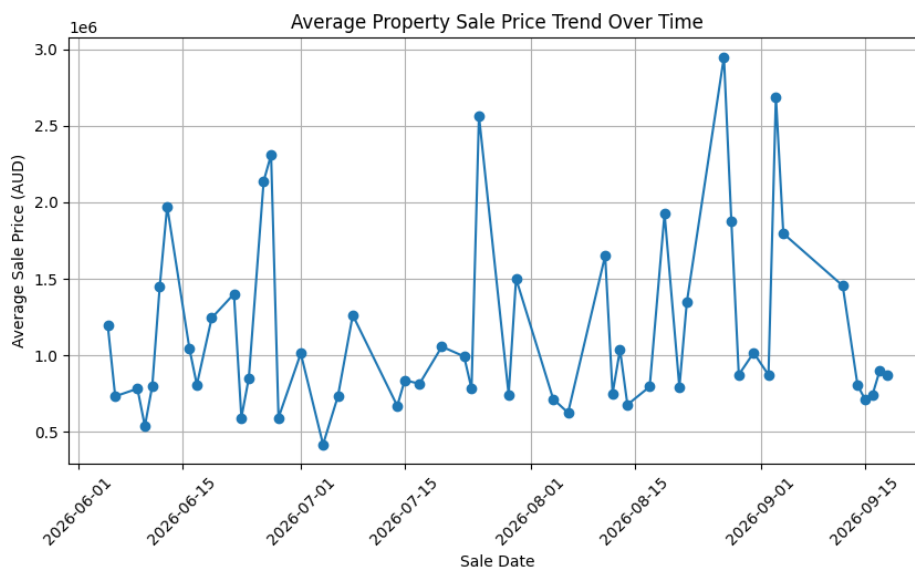


Figure 10: Average property sale price trend over time

2.6 Selection of Important Variables for Housing Price Prediction

Before creating engineered features, three variables were identified as potentially having the strongest influence on housing prices.

1. Market Region

Location is one of the most important determinants of housing value. The selected markets represent different levels of accessibility, demand, and property value. Therefore, market region is expected to have a strong influence on sale price.

2. Property Type

Different property types, such as houses and apartments, generally have different pricing patterns due to variations in land availability, size, and buyer demand.

3. Number of Bedrooms

The number of bedrooms represents property size and residential capacity. Larger properties generally have higher market values due to increased usable space.

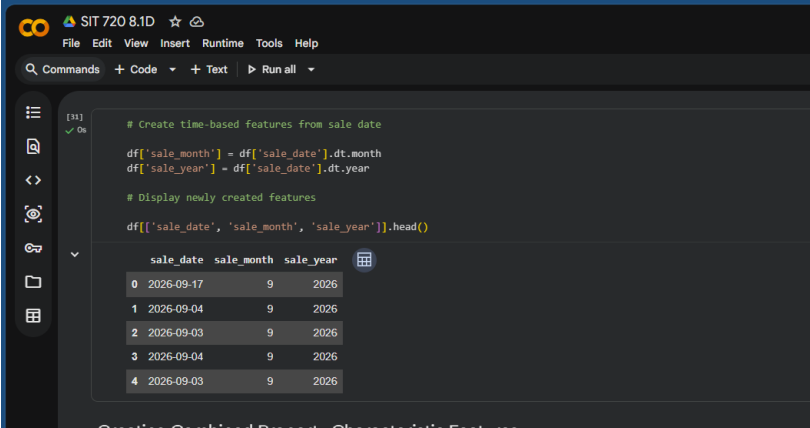
These variable selections align with the initial understanding of the housing market because geographic, structural, and size-related characteristics are major contributors to property prices.

2.7 Feature Engineering

Additional features were created from the existing property attributes to improve the representation of factors influencing housing prices.

2.7.1 Time-Based Features

Sale date information was transformed into separate variables representing sale month and sale year. These features allow future predictive models to capture possible temporal patterns in property transactions.



```
[11]: # Create time-based features from sale date
df['sale_month'] = df['sale_date'].dt.month
df['sale_year'] = df['sale_date'].dt.year

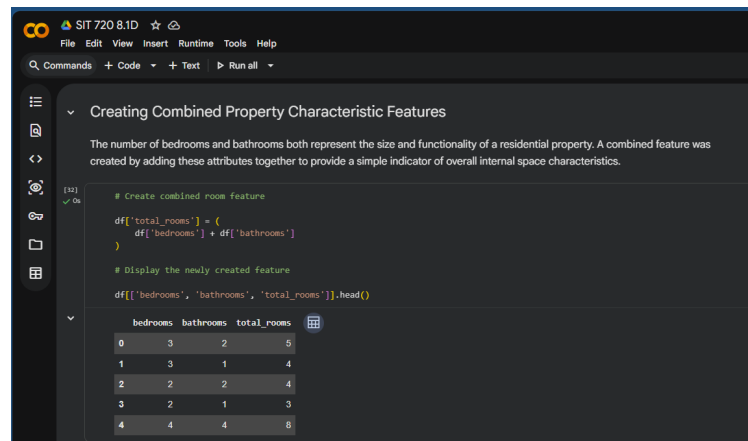
# Display newly created features
df[['sale_date', 'sale_month', 'sale_year']].head()
```

	sale_date	sale_month	sale_year
0	2026-09-17	9	2026
1	2026-09-04	9	2026
2	2026-09-03	9	2026
3	2026-09-04	9	2026
4	2026-09-03	9	2026

Figure 11: Creation of time-based features

2.7.2 Combined Property Characteristics

A new feature called *total rooms* was created by combining bedrooms and bathrooms. This feature provides a simple representation of the overall internal space characteristics of each property.



The screenshot shows a Jupyter Notebook titled "SIT 720 8.1D". The code cell is titled "Creating Combined Property Characteristic Features". It contains the following code:

```
# Create combined room feature
df['total_rooms'] = (
    df['bedrooms'] + df['bathrooms']
)

# Display the newly created feature
df[['bedrooms', 'bathrooms', 'total_rooms']].head()
```

The output shows a table with 5 rows and 3 columns: bedrooms, bathrooms, and total_rooms.

	bedrooms	bathrooms	total_rooms
0	3	2	5
1	3	1	4
2	2	2	4
3	2	1	3
4	4	4	8

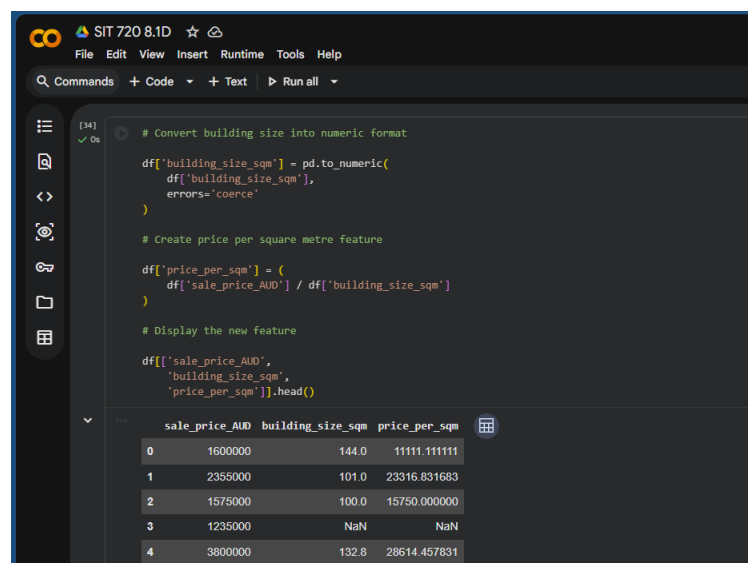
Figure 12: Creation of combined property feature

2.7.3 Size-Adjusted Price Feature

A size-adjusted feature named *price per square metre* was created using:

$$Price_per_sqm = \frac{Sale\ Price\ (AUD)}{Building\ Size\ (sqm)}$$

This feature represents property value relative to its size and may provide useful information for predictive modelling.



The screenshot shows a Jupyter Notebook titled "SIT 720 8.1D". The code cell is titled "Convert building size into numeric format". It contains the following code:

```
# Convert building size into numeric format
df['building_size_sqm'] = pd.to_numeric(
    df['building_size_sqm'],
    errors='coerce'
)

# Create price per square metre feature
df['price_per_sqm'] = (
    df['sale_price_AUD'] / df['building_size_sqm']
)

# Display the new feature
df[['sale_price_AUD',
    'building_size_sqm',
    'price_per_sqm']].head()
```

The output shows a table with 5 rows and 3 columns: sale_price_AUD, building_size_sqm, and price_per_sqm.

	sale_price_AUD	building_size_sqm	price_per_sqm
0	1600000	144.0	11111.111111
1	2355000	101.0	23316.831683
2	1575000	100.0	15750.000000
3	1235000	NaN	NaN
4	3800000	132.8	28614.457831

Figure 13: Creation of price per square metre feature

2.8 Reflection on Feature Engineering Decisions

The engineered features are consistent with the initial understanding of the Sydney housing market. Location-related variables capture differences between housing markets, while structural features represent property size and functionality.

The additional engineered variables provide further information regarding temporal effects, internal property characteristics, and size-adjusted valuation. These features will be considered during predictive modelling to improve the ability of machine learning models to estimate housing prices.

3. Part 3: Model Development and Evaluation

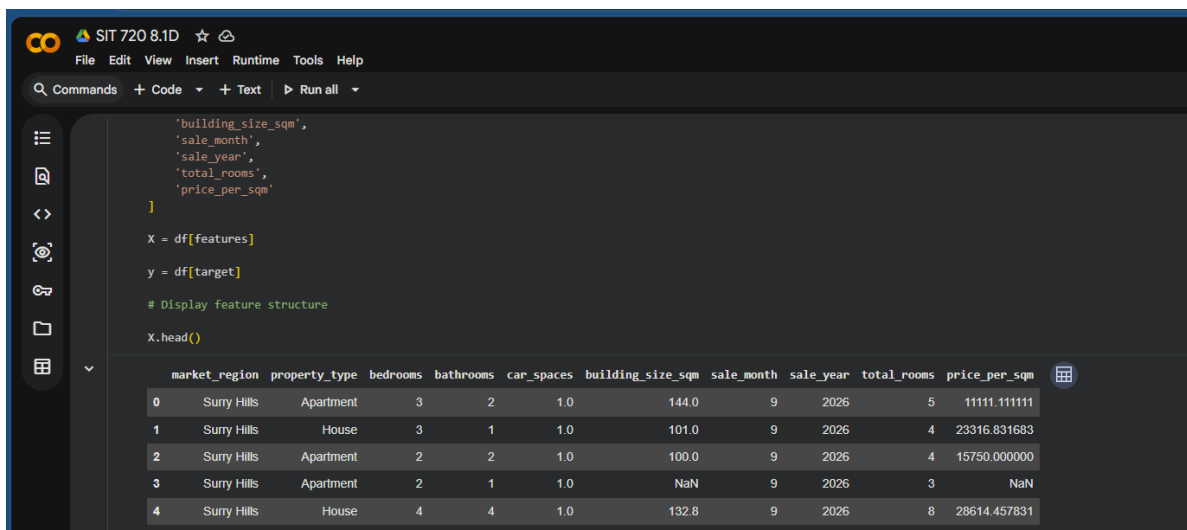
Three regression models were developed to predict residential property sale prices. The selected models represent different modelling approaches, including a simple linear baseline model and two ensemble learning approaches capable of capturing complex relationships within the housing market.

Before training the models, the target variable was defined as `sale_price_AUD`. The selected explanatory variables included market region, property type, bedrooms, bathrooms, car spaces, building size, sale month, sale year, total rooms, and price per square metre.

Categorical variables were transformed using one-hot encoding, while missing values were handled using appropriate imputation methods. This preprocessing step ensured that all models received properly formatted input data.

3.1 Data Preparation for Regression Models

The dataset was prepared by separating the target variable from the predictor variables. The selected features represent important factors influencing residential property prices, including location, property characteristics, and size-related attributes.



The screenshot shows a Jupyter Notebook interface with the following code and output:

```

'building_size_sqm',
'sale_month',
'sale_year',
'total_rooms',
'price_per_sqm'
]

X = df[features]
y = df[target]

# Display feature structure
X.head()

```

The output displays the first five rows of the feature matrix X:

	market_region	property_type	bedrooms	bathrooms	car_spaces	building_size_sqm	sale_month	sale_year	total_rooms	price_per_sqm
0	Surry Hills	Apartment	3	2	1.0	144.0	9	2026	5	11111.111111
1	Surry Hills	House	3	1	1.0	101.0	9	2026	4	23316.831683
2	Surry Hills	Apartment	2	2	1.0	100.0	9	2026	4	15750.000000
3	Surry Hills	Apartment	2	1	1.0	NaN	9	2026	3	NaN
4	Surry Hills	House	4	4	1.0	132.8	9	2026	8	28614.457831

Figure 14: Selected features and target variable preparation for regression modelling

3.2 Data Preprocessing

The dataset contained both numerical and categorical variables. Since regression algorithms cannot directly process categorical attributes, categorical variables were converted into numerical representations using one-hot encoding.

Missing numerical values were handled using median imputation, while missing categorical values were replaced using the most frequent category. A preprocessing pipeline was created using a `ColumnTransformer` to apply the required transformations.

The dataset contains both numerical and categorical variables.

Categorical variables such as housing market region and property type cannot be directly used by regression algorithms. Therefore, one-hot encoding is applied to transform categorical variables into numerical representations.

Numerical variables are passed without transformation.

```
# Identify numerical and categorical features

categorical_features = [
    'market_region',
    'property_type'
]

numerical_features = [
    'bedrooms',
    'bathrooms',
    'car_spaces',
    'building_size_sqm',
    'sale_month',
    'sale_year',
    'total_rooms',
    'price_per_sqm'
]

# Create preprocessing pipeline with missing value handling

from sklearn.impute import SimpleImputer

preprocessor = ColumnTransformer(
    transformers=[
        (
            'cat',
            Pipeline(
                steps=[
                    ('imputer', SimpleImputer(strategy='most_frequent')),
                    ('encoder', OneHotEncoder(handle_unknown='ignore'))
                ]
            ),
            categorical_features
        ),
        (
            'num',
            Pipeline(
                steps=[
                    ('imputer', SimpleImputer(strategy='median'))
                ]
            ),
            numerical_features
        )
    ]
)

preprocessor
```

Visual diagram of the ColumnTransformer structure:

```

graph LR
    subgraph ColumnTransformer
        direction LR
        subgraph cat
            imputer_cat[SimpleImputer] --> encoder_cat[OneHotEncoder]
        end
        subgraph num
            imputer_num[SimpleImputer]
        end
    end

```

Figure 15: Data preprocessing pipeline including encoding and missing value handling

3.3 Selection of Regression Models

Three regression approaches were selected to represent different modelling strategies.

Linear Regression

Linear Regression was selected as a baseline model because it provides a simple and interpretable relationship between input variables and property prices. However, it assumes linear relationships and may not capture complex interactions between housing features.

Random Forest Regression

Random Forest Regression was selected as an ensemble tree-based model. It combines multiple decision trees and can capture nonlinear relationships between variables such as location, property type, and size characteristics.

Gradient Boosting Regression

Gradient Boosting Regression was selected as a sequential ensemble approach where each new tree improves the errors made by previous trees. This allows the model to learn complex patterns in structured datasets.

Based on the characteristics of the housing dataset, ensemble models were expected to perform better than Linear Regression because property prices are influenced by multiple interacting factors.

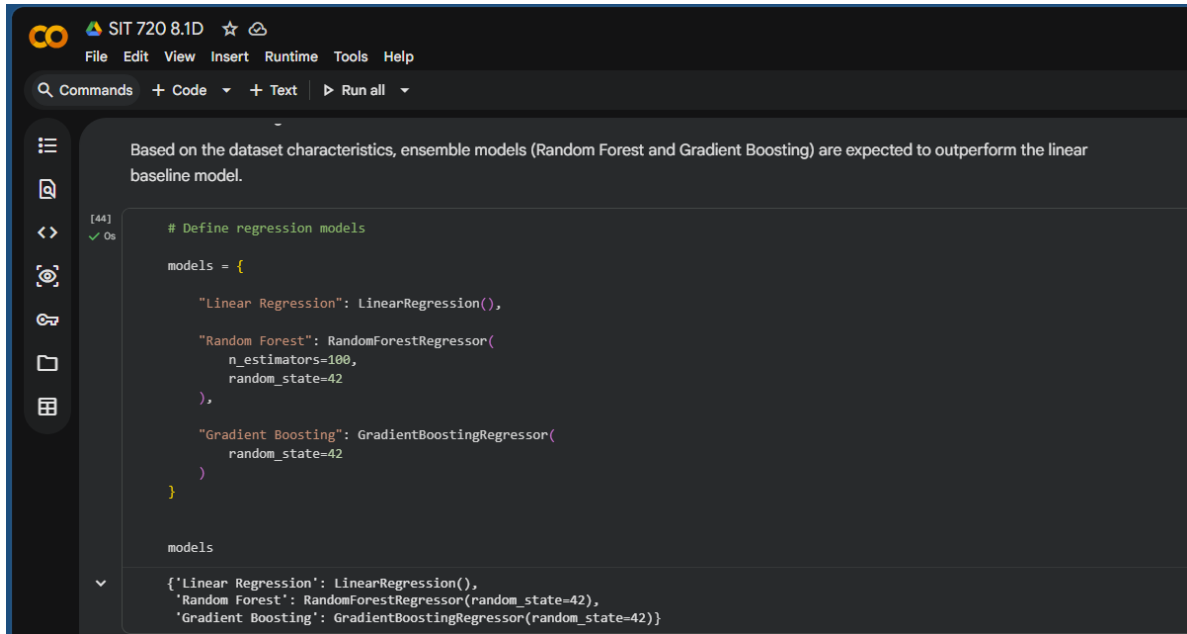


Figure 16: Regression models implemented in the study

3.4 K-Fold Cross-Validation and Evaluation Metrics

To provide a reliable comparison between models, 5-fold cross-validation was applied. The dataset was divided into five subsets, where each model was trained on four subsets and evaluated on the remaining subset. The process was repeated five times and the average performance was calculated.

The following evaluation metrics were used:

- Mean Absolute Error (MAE): Measures the average difference between predicted and actual property prices.
- Root Mean Squared Error (RMSE): Penalises larger prediction errors.
- R^2 Score: Measures how well the model explains variation in property prices.

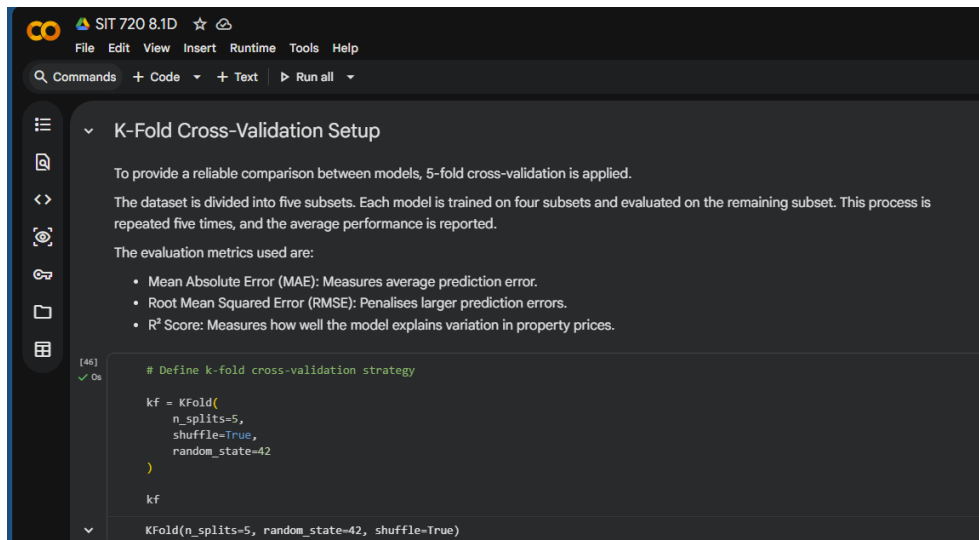


Figure 17: Five-fold cross-validation setup and evaluation metrics

3.5 Model Performance Comparison

The models were evaluated using the same 5-fold cross-validation strategy. The results are presented below.

Table 1: Regression model performance comparison

Model	MAE (AUD)	RMSE (AUD)	R ²
Linear Regression	272245.91	399625.02	0.480
Random Forest	212114.98	332427.54	0.662
Gradient Boosting	216455.51	334103.10	0.657

Random Forest achieved the best validation performance with the lowest MAE and RMSE and the highest R² score. Gradient Boosting also achieved similar performance, while Linear Regression showed comparatively weaker predictive ability.

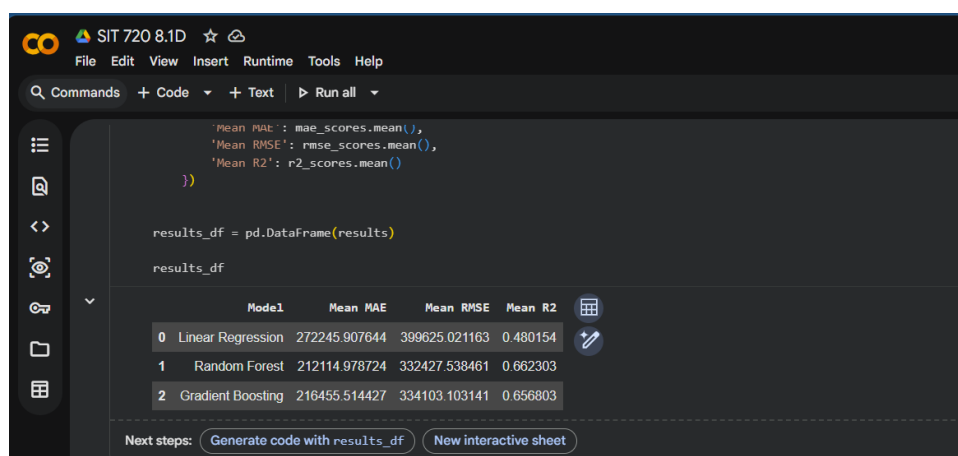


Figure 18: Cross-validation results for the three regression models

3.6 Model Behaviour Analysis

The difference between training and validation performance was analysed to identify possible underfitting or overfitting behaviour.

Linear Regression achieved a training R^2 value of approximately 0.761 and a lower validation R^2 value of 0.480. This indicates that the model was unable to capture the complex nonlinear relationships present within the housing dataset.

The ensemble models achieved higher training performance. Random Forest achieved a training R^2 value of approximately 0.956, while Gradient Boosting achieved approximately 0.975. The difference between training and validation results suggests some level of overfitting due to increased model complexity.

However, both ensemble approaches maintained considerably higher validation performance than Linear Regression, demonstrating better generalisation capability.

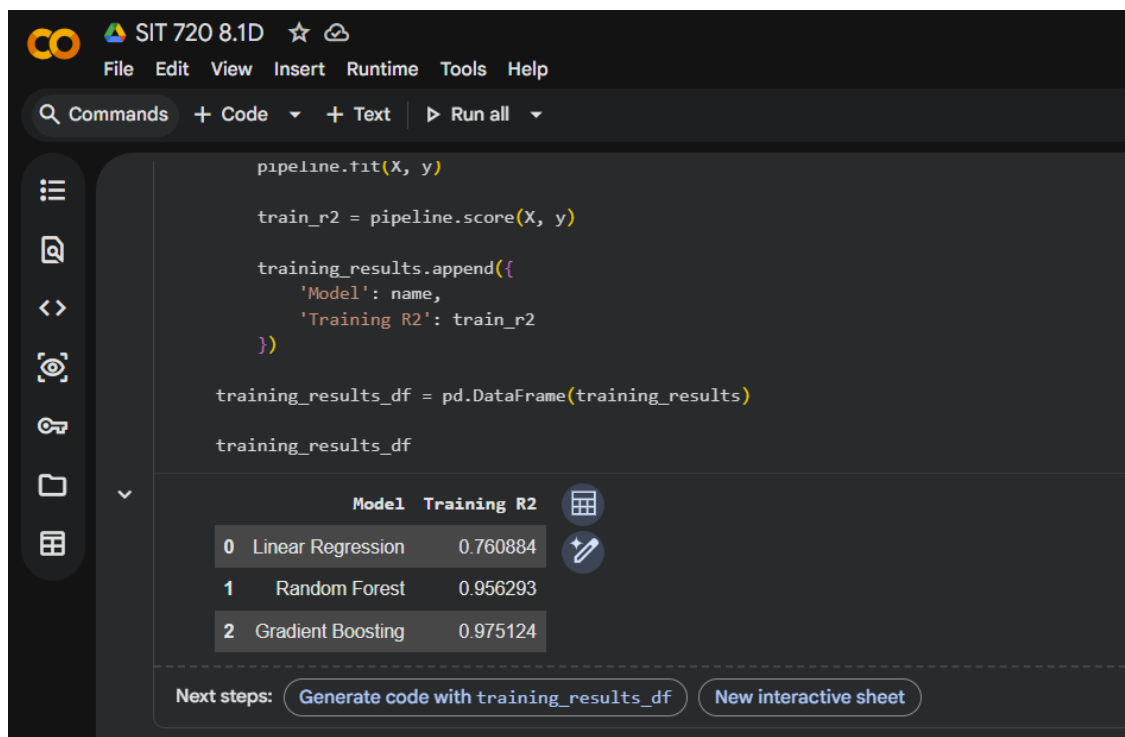


Figure 19: Training performance comparison between regression models

3.7 Revisiting Initial Expectations and Recommended Model

Before model development, it was expected that ensemble models would outperform the linear baseline because housing prices generally depend on nonlinear relationships between location, property characteristics, and size-related features.

The experimental results supported this expectation. Both Random Forest and Gradient Boosting achieved higher predictive performance compared with Linear Regression.

Based on the cross-validation results, Random Forest Regression was selected as the most appropriate model for predicting housing sale prices. It achieved the highest R^2 score and the lowest prediction errors among the tested models. Although it showed increased complexity compared with the linear model, its improved validation performance indicates better ability to capture meaningful patterns in the housing dataset.

4. Part 4: Investigating Prediction Failures

Machine learning models often reveal their limitations through cases where prediction errors are large. In this section, the largest prediction errors produced by the Random Forest regression model were identified and investigated. These cases were analysed to understand why the model failed and what additional information could improve future prediction performance.

4.1 Identification of Largest Prediction Errors

The Random Forest regression model was selected as the final predictive model based on its superior cross-validation performance compared with Linear Regression and Gradient Boosting. The trained model was used to generate predictions for the property sale prices. The predicted values were compared with the actual sale prices to identify cases with large prediction errors.

Figure 20 presents the initial comparison between actual property sale prices and the predicted values generated by the Random Forest model.

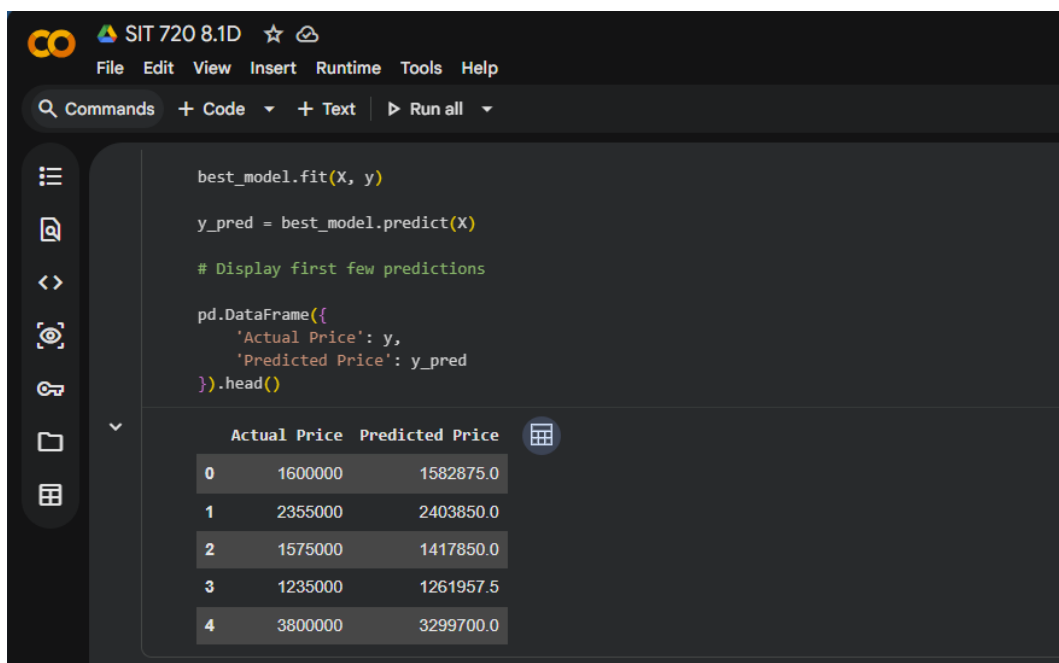


Figure 20: Actual and predicted property sale prices using Random Forest

The prediction error was calculated for each property, and the five properties with the largest absolute errors were selected for detailed investigation. These properties represent cases where the model was unable to accurately estimate the market value.

Figure 21 shows the properties producing the largest prediction errors.

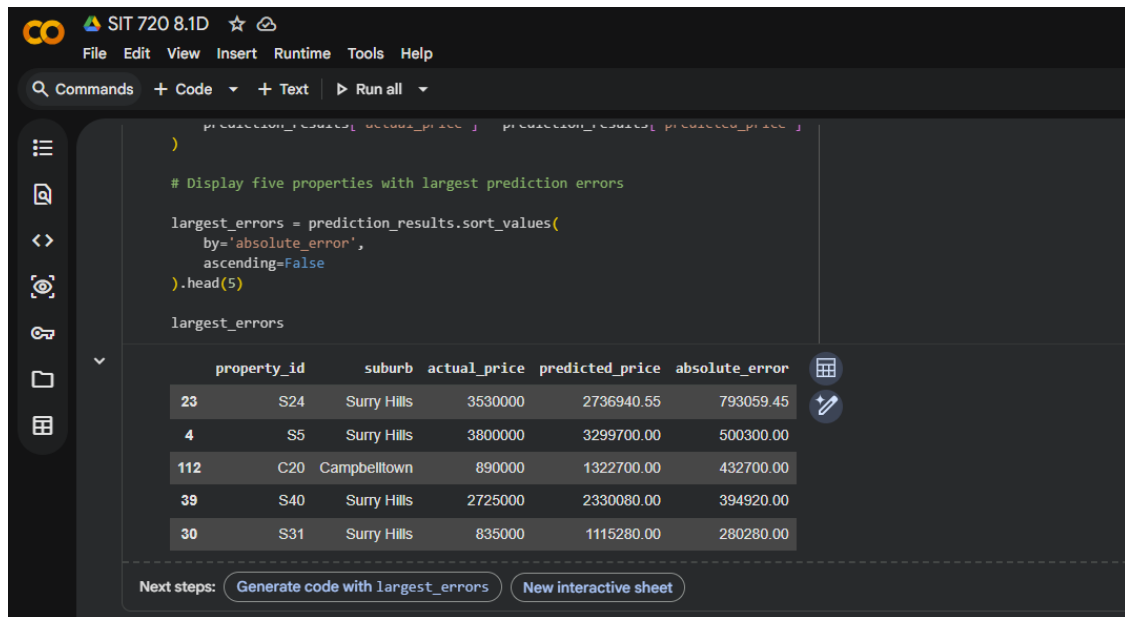


Figure 21: Five properties with the largest prediction errors

4.2 Investigation of Prediction Failure Cases

The characteristics of the selected properties were examined to understand possible reasons behind the prediction failures. Figure 22 presents the main characteristics of these properties.

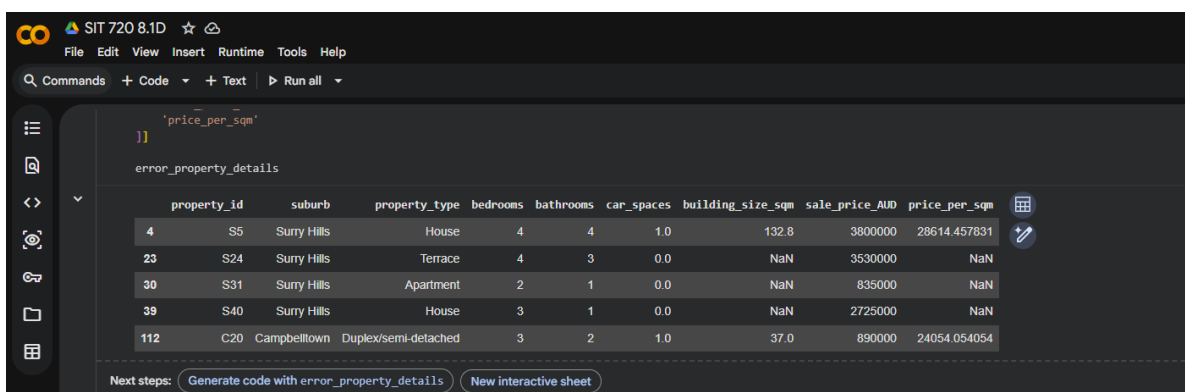


Figure 22: Characteristics of properties with the largest prediction errors

4.2.1 Analysis of High Error Properties

The largest prediction errors were mainly observed for properties located in Surry Hills and Campbelltown. Property S24 from Surry Hills produced one of the largest errors, where the actual sale price was substantially higher than the predicted value. Although the property contained favourable structural characteristics such as multiple bedrooms and bathrooms, the model underestimated its market value.

Similarly, property S5 from Surry Hills also showed a large difference between actual and predicted prices. The property had four bedrooms, four bathrooms, and a relatively large building size. However, premium properties often contain unique characteristics that are difficult to capture using only numerical and categorical variables.

Property C20 from Campbelltown showed the opposite pattern, where the model overestimated the sale price. The property was a duplex/semi-detached dwelling with three bedrooms and two bathrooms. This suggests

that the model learned general relationships between property features and prices but could not fully represent local market conditions.

Properties S40 and S31 also demonstrated substantial prediction errors. Both were located in Surry Hills, where property values can vary significantly depending on street location, accessibility, property condition, and other neighbourhood-level factors. These variables were not available in the dataset and therefore could not be considered by the model.

4.3 Limitations of the Modelling Approach

The prediction failures highlight several limitations of the current modelling approach. Although the Random Forest model successfully captured general patterns between property characteristics and sale prices, some important factors influencing housing values were unavailable.

The dataset mainly contained structural information such as property type, bedrooms, bathrooms, car spaces, building size, suburb, and price per square metre. However, housing prices are also affected by additional factors including:

- exact geographic location;
- land size and land value;
- distance to public transport;
- proximity to schools and amenities;
- renovation quality and property condition;
- building age;
- views and environmental characteristics;
- current market demand.

The absence of these variables limits the ability of machine learning models to accurately predict individual property prices, especially for unique or high-value properties.

4.4 Reflection on Difficult-to-Model Properties

The analysis suggests that some property categories are inherently more difficult to model than others. Standard residential properties with common characteristics are generally easier to predict because similar examples exist within the training data.

However, premium properties and properties located in expensive suburbs such as Surry Hills are more challenging. Small differences in location, architectural design, renovation quality, and surrounding environment can create large differences in market value. These characteristics are difficult to represent using limited features.

Future prediction performance could be improved by including additional data sources such as geographic coordinates, land information, transport accessibility, school zones, neighbourhood statistics, and property condition indicators.

Therefore, predictions from the current model should be interpreted cautiously, particularly for unusual properties, luxury homes, and cases where important contextual information is unavailable.

5. Part 5: Human Judgement, Machine Learning, and Large Language Models

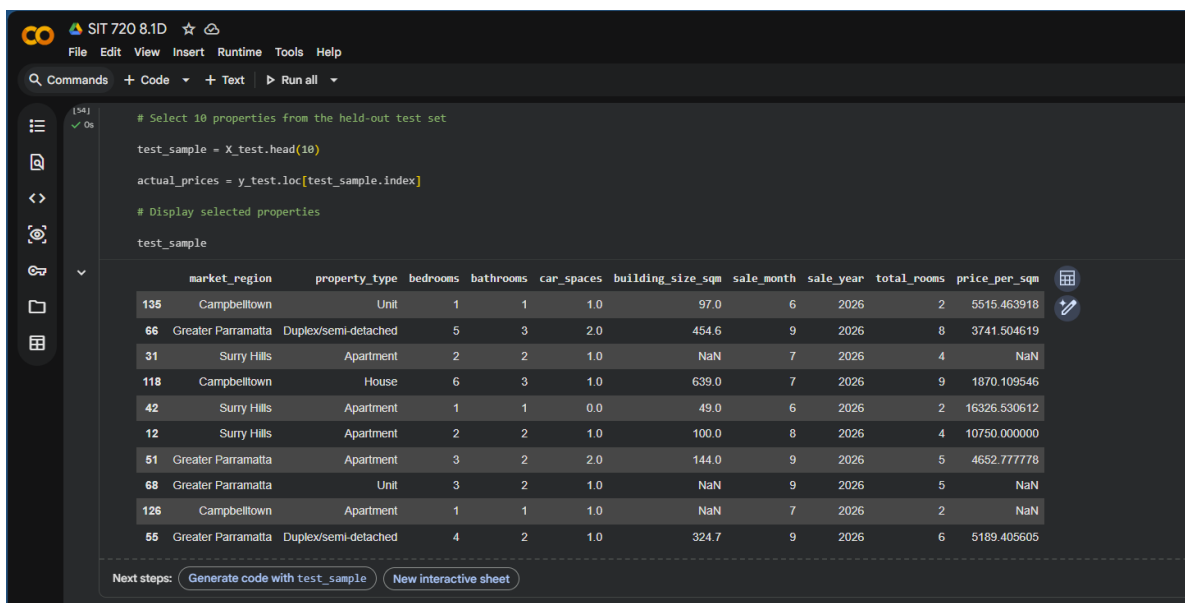
This section compares three different approaches for residential property valuation: (1) machine learning prediction using the best-performing model identified in Part 3, (2) large language model (LLM)-based estimation, and (3) human judgement-based estimation using housing market understanding developed throughout the project.

A held-out test set was used to ensure that the comparison was performed on unseen properties. Ten properties were selected from the test dataset, and their actual sale prices were compared against predictions generated by the three valuation approaches.

5.1 Selection of Held-out Test Properties

The dataset was divided into training and testing subsets using an 80:20 split. Ten properties were selected from the held-out test set for comparison. These properties represented different suburbs, including Surry Hills, Campbelltown, and Greater Parramatta, with varying property types, numbers of bedrooms, bathrooms, car spaces, and building sizes.

The selected properties contained diverse characteristics, allowing evaluation of how different valuation approaches perform across different housing scenarios.



```
# Select 10 properties from the held-out test set
test_sample = X_test.head(10)

actual_prices = y_test.loc[test_sample.index]

# Display selected properties
test_sample
```

	market_region	property_type	bedrooms	bathrooms	car_spaces	building_size_sqm	sale_month	sale_year	total_rooms	price_per_sqm
135	Campbelltown	Unit	1	1	1.0	97.0	6	2026	2	5515.463918
66	Greater Parramatta	Duplex/semi-detached	5	3	2.0	454.6	9	2026	8	3741.504619
31	Surry Hills	Apartment	2	2	1.0	NaN	7	2026	4	NaN
118	Campbelltown	House	6	3	1.0	639.0	7	2026	9	1870.109546
42	Surry Hills	Apartment	1	1	0.0	49.0	6	2026	2	16326.530612
12	Surry Hills	Apartment	2	2	1.0	100.0	8	2026	4	10750.000000
51	Greater Parramatta	Apartment	3	2	2.0	144.0	9	2026	5	4652.777778
68	Greater Parramatta	Unit	3	2	1.0	NaN	9	2026	5	NaN
126	Campbelltown	Apartment	1	1	1.0	NaN	7	2026	2	NaN
55	Greater Parramatta	Duplex/semi-detached	4	2	1.0	324.7	9	2026	6	5189.405605

Figure 23: Selected properties from the held-out test set

5.2 Machine Learning-Based Property Valuation

Based on the evaluation results from Part 3, Random Forest Regression was selected as the best-performing machine learning model. The model achieved the highest cross-validation performance compared with Linear Regression and Gradient Boosting Regression.

The Random Forest model was trained using the training dataset and used to predict sale prices for the ten selected test properties.

The predictions were compared with the actual sale prices. The machine learning model was able to capture general relationships between property characteristics and price patterns; however, some differences occurred

for properties with unusual characteristics or missing information.

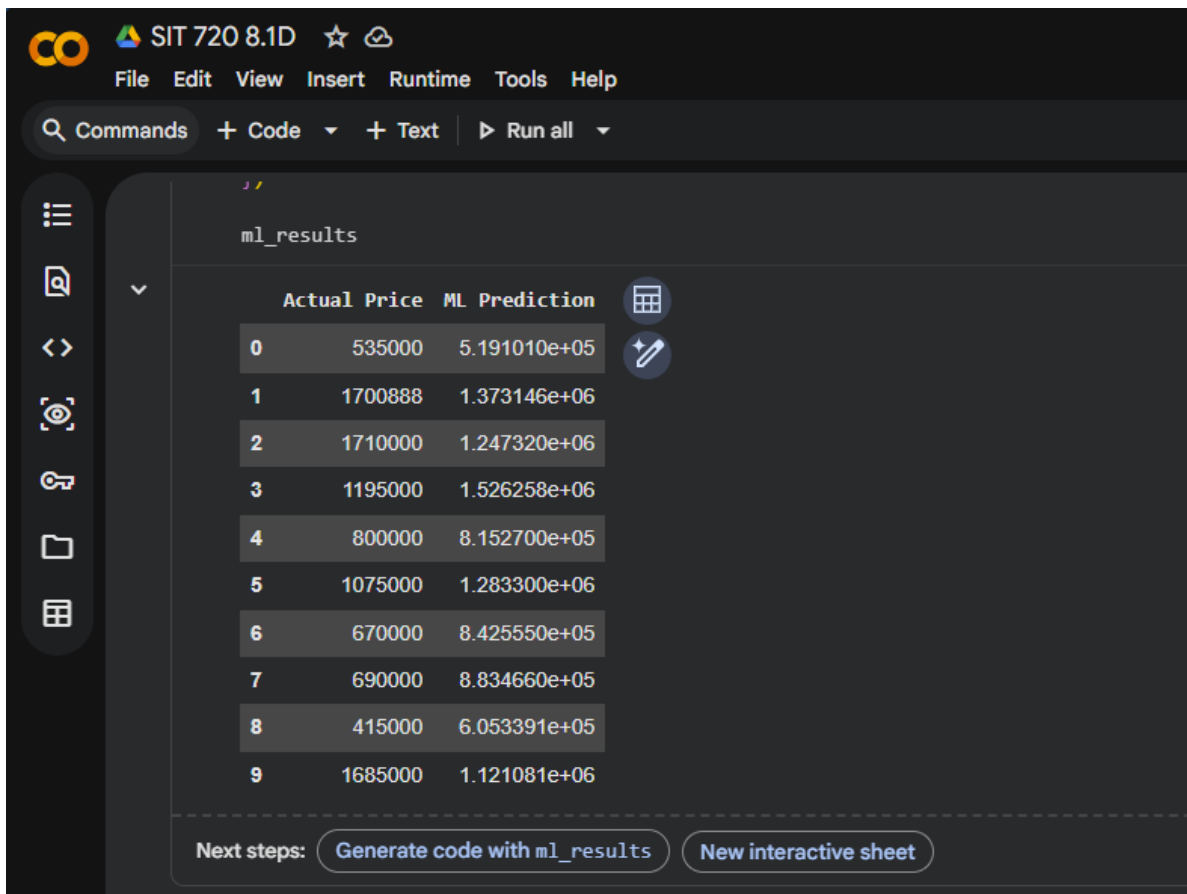


Figure 24: Machine learning predictions generated using Random Forest Regression

5.3 Large Language Model (LLM)-Based Estimation

The second approach used a large language model-based estimation strategy. The LLM estimates were generated using the same property characteristics available to the machine learning model, including suburb, property type, bedrooms, bathrooms, car spaces, and building size.

Unlike machine learning models that learn patterns from historical data, an LLM relies on contextual understanding and general knowledge about housing markets. Therefore, LLM predictions may consider qualitative relationships between property characteristics and expected market value.

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[37] On

Add columns for LLM and human estimates

comparison_table["LLM Prediction"] = None

comparison_table["Human Prediction"] = None

comparison_table

	index	market_region	property_type	bedrooms	bathrooms	car_spaces	building_size_sqm	sale_month	sale_year	total_rooms	price_per_sqm	Actual Price	ML Prediction	LLM Prediction	Human Prediction
0	135	Campbelltown	Unit	1	1	1.0	97.0	6	2026	2	5515.463918	535000	5.191010e+05	None	None
1	66	Greater Paramatta	Duplex/semi-detached	5	3	2.0	454.6	9	2026	8	3741.504619	1700888	1.373146e+06	None	None
2	31	Surry Hills	Apartment	2	2	1.0	NaN	7	2026	4	NaN	1710000	1.247320e+06	None	None
3	118	Campbelltown	House	6	3	1.0	639.0	7	2026	9	1870.109546	1195000	1.526258e+06	None	None
4	42	Surry Hills	Apartment	1	1	0.0	49.0	6	2026	2	16326.530612	800000	8.152700e+05	None	None
5	12	Surry Hills	Apartment	2	2	1.0	100.0	8	2026	4	10750.000000	1075000	1.283300e+06	None	None
6	51	Greater Paramatta	Apartment	3	2	2.0	144.0	9	2026	5	4652.777778	670000	8.425550e+05	None	None
7	68	Greater Paramatta	Unit	3	2	1.0	NaN	9	2026	5	NaN	690000	8.834660e+05	None	None
8	126	Campbelltown	Apartment	1	1	1.0	NaN	7	2026	2	NaN	415000	6.053391e+05	None	None
9	55	Greater Paramatta	Duplex/semi-detached	4	2	1.0	324.7	9	2026	6	5189.405605	1685000	1.121081e+06	None	None

Figure 25: Comparison table containing actual prices, machine learning predictions, and LLM predictions

5.4 Human Judgement-Based Estimation

The third approach involved manual estimation based on housing market knowledge and insights obtained during the project. Human estimates considered factors such as suburb desirability, property type, number of bedrooms, building size, and expected market demand.

Human judgement can incorporate contextual factors that are difficult to represent numerically, such as neighbourhood reputation, property appeal, and market trends. However, human estimates may also be affected by personal assumptions and limited access to complete market information.

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```
comparison_table["LLM Prediction"] = llm_predictions

comparison_table
```

	index	market_region	property_type	bedrooms	bathrooms	car_spaces	building_size_sqm	sale_month	sale_year	total_rooms	price_per_sqm	Actual Price	ML Prediction	LLM Prediction	Human Prediction
0	135	Campbelltown	Unit	1	1	1.0	97.0	6	2025	2	5515.463918	535000	5.191010e+05	550000	None
1	66	Greater Parramatta	Duplex/semi-detached	5	3	2.0	454.6	9	2025	8	3741.504619	1700688	1.373146e+06	1600000	None
2	31	Surry Hills	Apartment	2	2	1.0	NaN	7	2025	4	NaN	1710000	1.247320e+05	1400000	None
3	118	Campbelltown	House	6	3	1.0	639.0	7	2026	9	1870.109546	1195000	1.526258e+06	1500000	None
4	42	Surry Hills	Apartment	1	1	0.0	49.0	6	2025	2	16326.530612	800000	8.152700e+05	750000	None
5	12	Surry Hills	Apartment	2	2	1.0	100.0	8	2025	4	10750.000000	1075000	1.283300e+06	1100000	None
6	51	Greater Parramatta	Apartment	3	2	2.0	144.0	9	2025	5	4652.777778	670000	8.425550e+05	800000	None
7	68	Greater Parramatta	Unit	3	2	1.0	NaN	9	2026	5	NaN	690000	8.834660e+05	850000	None
8	126	Campbelltown	Apartment	1	1	1.0	NaN	7	2026	2	NaN	415000	6.053391e+05	450000	None
9	55	Greater Parramatta	Duplex/semi-detached	4	2	1.0	324.7	9	2025	6	5189.405605	1685000	1.121081e+06	1500000	None

Figure 26: Final comparison of actual prices, ML predictions, LLM predictions, and human judgement estimates

5.5 Comparison of Prediction Errors

To quantitatively compare the three approaches, Mean Absolute Error (MAE) was calculated. A lower MAE indicates that the predictions are closer to the actual sale prices.

The obtained results are shown below:

Table 2: Mean Absolute Error comparison between valuation approaches

Approach	Mean Absolute Error (AUD)
Machine Learning (Random Forest)	248142.81
LLM	131588.80
Human Judgement	86411.20

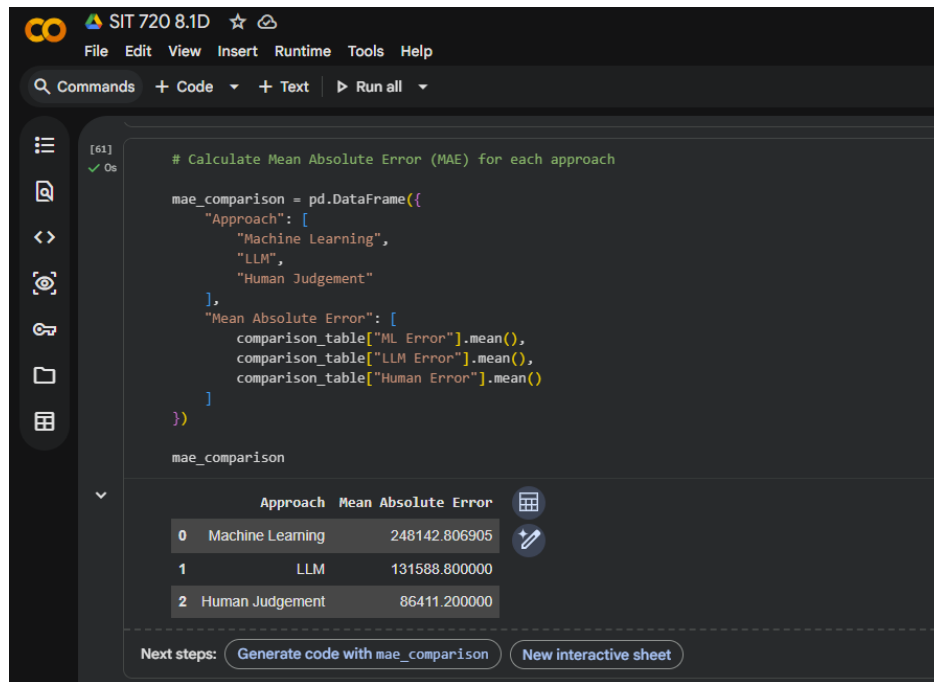


Figure 27: Mean Absolute Error comparison of three valuation approaches

5.6 Critical Discussion

The comparison demonstrates that all three approaches were capable of producing reasonable property value estimates, but their prediction behaviour differed considerably.

The machine learning model provided a systematic and data-driven valuation approach. Random Forest was able to identify complex relationships between property features and historical sale prices. However, its performance was limited when important influencing factors were unavailable in the dataset, such as exact location, land value, renovation quality, property condition, and local market changes.

The LLM approach produced lower prediction errors than the machine learning model for the selected properties. This suggests that contextual understanding and general market knowledge can provide additional value when numerical information is limited. However, LLM-based estimation may lack consistency because predictions depend on the information provided and the model's interpretation of market conditions.

Human judgement achieved the lowest MAE among the three approaches in this experiment. This indicates that human knowledge of housing markets and contextual factors can still contribute value in individual property valuation tasks. Humans can consider qualitative characteristics that are difficult to encode in a dataset, such as street appeal, neighbourhood perception, and property uniqueness.

However, human judgement is also subjective and difficult to scale. Unlike machine learning models, human valuation requires time and expertise for each individual property. Therefore, automated models remain valuable for large-scale prediction tasks.

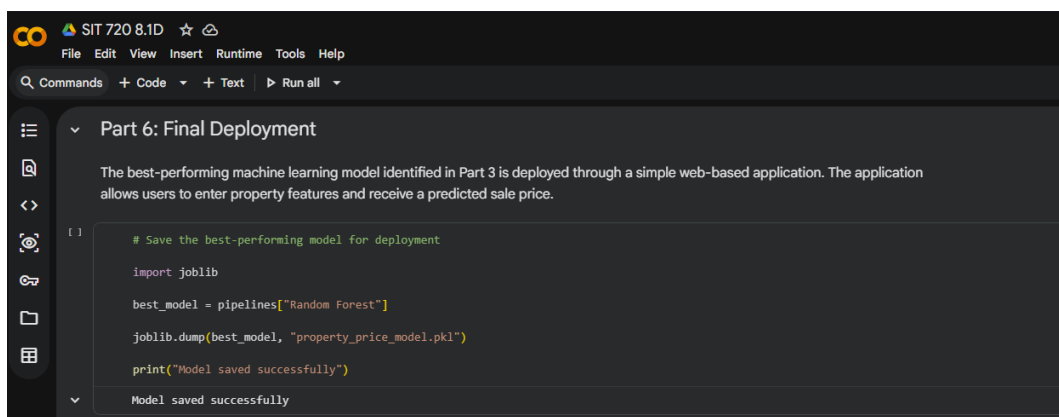
Overall, the results suggest that machine learning models provide efficient and reproducible predictions, while LLMs and human judgement can provide additional contextual interpretation. A combined approach integrating data-driven models with expert knowledge may provide improved property valuation performance in real-world applications.

6. Part 6: Final Deployment and Reflection

The final stage of this project involved deploying the selected machine learning model into a practical web-based application. Based on the evaluation results from Part 3, the Random Forest Regression model was selected as the final model due to its superior predictive performance compared with the other evaluated regression approaches. The objective of deployment was to allow users to provide property characteristics and obtain an estimated sale price generated by the trained model.

6.1 Final Model Deployment

The trained Random Forest pipeline was saved using the Joblib library to enable reuse of the developed model without retraining. The saved model included the complete preprocessing pipeline and prediction model, allowing new property information to be processed consistently during deployment.



```
[ ]  
# Save the best-performing model for deployment  
  
import joblib  
  
best_model = pipelines["Random Forest"]  
  
joblib.dump(best_model, "property_price_model.pkl")  
  
print("Model saved successfully")  
  
Model saved successfully
```

Figure 28: Saved Random Forest model for deployment

6.2 Web Application Development

A Streamlit-based web application was developed to provide an interactive interface for property price prediction. The application loads the previously saved Random Forest model and allows users to enter property characteristics, including market region, property type, number of bedrooms, bathrooms, car spaces, building size, sale month, sale year, and price per square metre.

The application was developed using Python libraries including Streamlit, Pandas, and Joblib. Streamlit was selected because it provides a simple framework for transforming machine learning models into interactive applications without requiring complex web development frameworks.

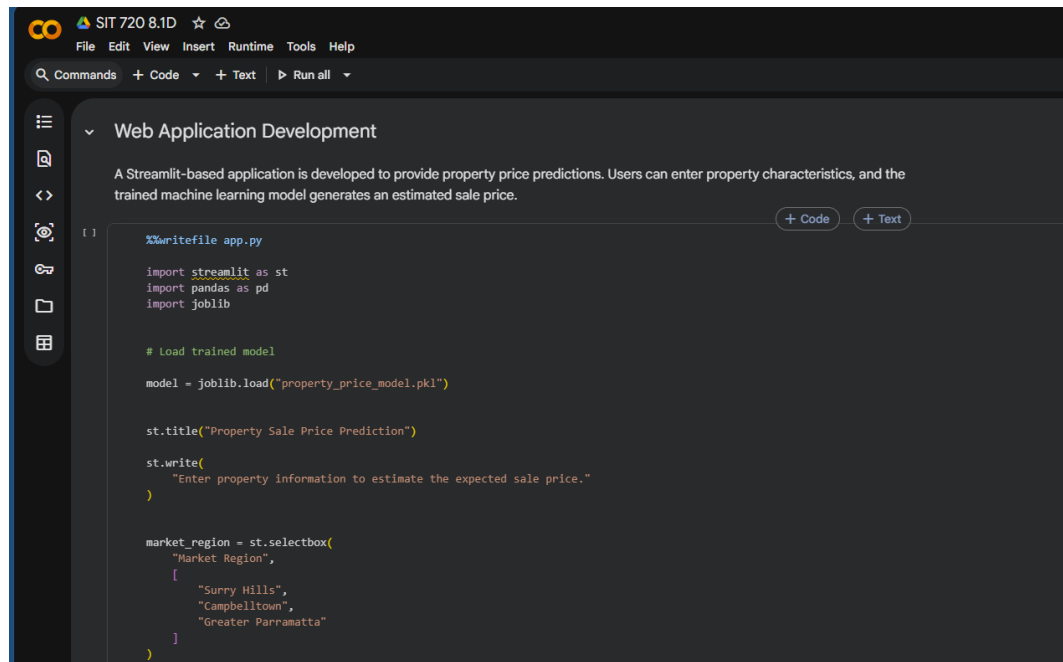


Figure 29: Streamlit application development using the deployed Random Forest model

6.3 Application Usage

The deployed application accepts user-defined property features and generates a predicted sale price using the trained machine learning model. Users can select property characteristics through the interactive interface and receive an immediate prediction result.

The application successfully generated a predicted property sale price after receiving user inputs, demonstrating the practical usability of the developed machine learning solution.

Application Usage Instructions:

The deployed Streamlit application can be used through the following steps:

1. Select the property market region and property type from the available options.
2. Enter property characteristics including the number of bedrooms, bathrooms, car spaces, and building size.
3. Provide additional sale-related information such as sale month, sale year, and price per square metre.
4. Click the “Predict Sale Price” button to generate the estimated property value.
5. The application displays the predicted sale price generated by the trained Random Forest model.

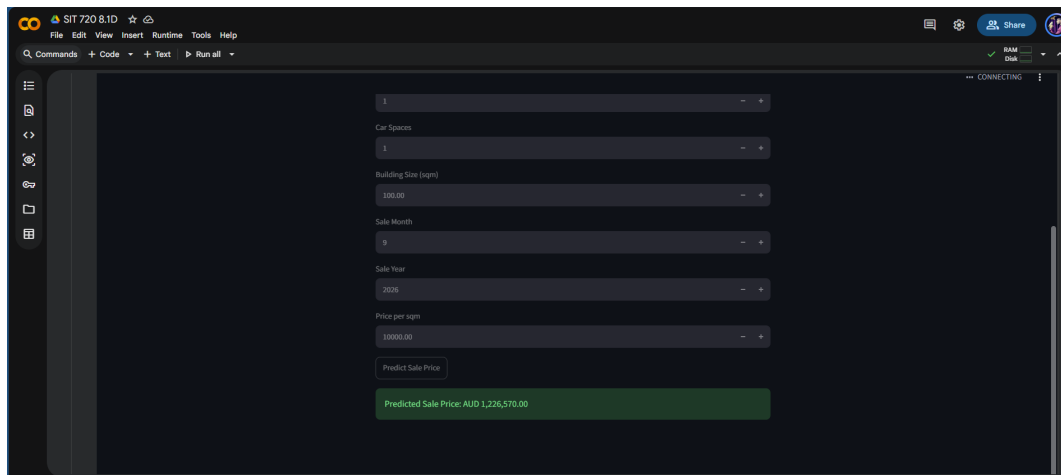


Figure 30: Predicted sale price generated by the deployed Streamlit application

6.4 Final Reflection on the Machine Learning Workflow

6.4.1 Key Lessons Learned

This project provided practical experience in developing a complete machine learning workflow, beginning with data preparation and ending with model deployment. The process demonstrated that successful machine learning applications require careful consideration of data quality, feature selection, preprocessing, model evaluation, and deployment requirements.

The project highlighted that predictive performance depends not only on algorithm selection but also on the availability and quality of relevant property information. A well-designed workflow is essential for transforming raw data into a reliable prediction system.

6.4.2 Challenges During Data Collection and Preprocessing

One of the major challenges was working with a relatively limited property dataset. Real estate prices are influenced by many factors, and some important variables were unavailable in the dataset. Missing values were present in some property attributes, requiring suitable preprocessing strategies before model development.

Categorical variables such as market region and property type required transformation before being used by machine learning algorithms. Numerical and categorical features were processed through a consistent pipeline to ensure reliable model training and prediction.

6.4.3 Feature Engineering Challenges

Feature engineering played an important role in improving the representation of property characteristics. Additional variables such as sale month, sale year, total rooms, and price per square metre were incorporated to provide additional information to the model.

However, the project also showed the limitations of feature engineering when important real-world information is unavailable. Factors such as property condition, renovation history, land size, accessibility, nearby facilities, and local market trends could potentially improve prediction accuracy.

6.4.4 Model Development and Evaluation

Three regression approaches were developed and evaluated during the project: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. The results showed that ensemble-based tree models performed better because property prices usually involve complex relationships between multiple factors.

The Random Forest model was selected for deployment because it provided strong predictive performance while maintaining practical usability. The evaluation process also demonstrated the importance of balancing model complexity, generalisation ability, and reliability on unseen data.

6.4.5 Trade-off Between Predictive Performance and Practical Deployment

Although more complex models may provide improved predictive performance, practical deployment requires consideration of computational requirements, maintenance, interpretability, and user accessibility.

The deployed Streamlit application provides a simple interface that allows users to obtain predictions without requiring technical knowledge. This demonstrates the importance of balancing predictive accuracy with practical implementation.

6.4.6 Ethical Implications of Automated Property Valuation

Automated property valuation systems may introduce bias if the training data does not adequately represent different property types, locations, or market conditions. Since machine learning models learn patterns from historical data, existing biases within historical transactions may influence future predictions.

Therefore, automated valuation systems should be used as decision-support tools rather than complete replacements for expert judgement. Transparency regarding model limitations and data constraints is important for responsible deployment.

6.4.7 Future Improvements

The project could be improved by collecting a larger and more diverse property dataset containing additional relevant variables. Including information such as land size, property condition, renovation history, transport accessibility, nearby facilities, and economic indicators could improve prediction performance.

Additional computational resources could also allow more advanced modelling approaches, extensive hyperparameter optimisation, and stronger validation strategies.

6.5 Conclusion

This project successfully developed an end-to-end machine learning solution for residential property price prediction, covering data preparation, feature engineering, model evaluation, and deployment. The Random Forest model provided an effective balance between predictive performance and practical usability.

However, the deployed prediction system should be considered a decision-support tool rather than a complete replacement for professional property valuation, as prediction accuracy depends on the quality, diversity, and completeness of the available dataset.

The final Streamlit application demonstrated how a trained machine learning model can be transformed into an accessible prediction tool. However, the project also highlighted the importance of considering data limitations, ethical issues, and real-world factors when applying automated valuation systems.

GenAI Acknowledgement

ChatGPT was used to support planning, code organisation, debugging, and improving the clarity of explanations. All generated suggestions were reviewed, tested, and modified before inclusion in this report. The final analysis and interpretation represent my own understanding of the task.